

VIRASORO COHOMOLOGY AND HAMILTONIAN REDUCTION

1.1 THE STATE SPACE AND THE FIRST ANOMALY

A degree-one operator can define cohomology only when its degree-two square vanishes on the specified state space. For the Virasoro algebra, this square is a quadratic ghost operator. Its coefficient comes from a finite contraction in the ghost representation. The zero modes then determine which restrictions form cochain complexes.

All vector spaces and tensor products below are algebraic over $\mathbb{C}$. For homogeneous operators, the bracket is $[A, B]_s = AB - (-1)^{|A||B|}BA$. Braces denote the bracket of two odd operators. Matter operators have ghost degree zero.

Let $\mathcal{M}$ be a Virasoro module with central charge $C \in \mathbb{C}$:

$$[L_m^M, L_n^M] = (m - n)L_{m+n}^M + \frac{C}{12}(m^3 - m)\delta_{m+n,0} \mathbf{1}. \quad (1.1)$$

Assume that $\mathcal{M}$ is restricted: for each $v \in \mathcal{M}$, $L_n^M v = 0$ for all sufficiently large positive n . No inner product or finite-dimensional weight-space hypothesis is imposed.

The ghost Clifford algebra has odd generators $b_n, c_n, n \in \mathbb{Z}$, and relations

$$\{b_m, c_n\} = \delta_{m+n,0}, \quad \{b_m, b_n\} = \{c_m, c_n\} = 0.$$

Its conformal vacuum Ω obeys

$$b_n \Omega = 0 \quad (n \geq -1), \quad c_n \Omega = 0 \quad (n \geq 2). \quad (1.2)$$

The ghost module is

$$\mathcal{F} = \Lambda(b_n \ (n \leq -2), c_n \ (n \leq 1))\Omega.$$

A displayed generator acts by exterior multiplication. Its Clifford partner acts by left exterior differentiation. Thus the formulas specify the signs on every monomial. Set $\deg \Omega = 0$, $\deg c_n = 1$, and $\deg b_n = -1$. Set $E\Omega = 0$ and assign both b_n and c_n energy $-n$. The energy of a monomial is the sum of these assignments.

Normal ordering, denoted by colons, moves the creators in (1.2) to the left. Each interchange of odd factors contributes a minus sign. Contraction terms are omitted. This convention applies to the whole monomial inside the colons. Define

$$L_m^{\text{gh}} = \sum_{s \in \mathbb{Z}} (m + s) : b_{m-s} c_s :, \quad (1.3)$$

$$R = -\frac{1}{2} \sum_{m,n \in \mathbb{Z}} (m - n) : c_{-m} c_{-n} b_{m+n} :, \quad (1.4)$$

$$Q = \sum_{n \in \mathbb{Z}} c_{-n} L_n^M + R \quad \text{on } \mathcal{D} = \mathcal{M} \otimes \mathcal{F}. \quad (1.5)$$

Tensor factors commute because the matter action is even. The degree of Q is one.

LEMMA 1.1.1 (*Algebraic domain*). Every sum in (1.3)–(1.5) has finitely many nonzero terms on each element of $\mathcal{D}$. The same holds for

$$\nu_r = \frac{1}{2} \sum_{a+b=r} (a-b)c_a c_b, \quad K = \frac{C-26}{12} \sum_{n>0} (n^3-n)c_{-n}c_n.$$

These operators and all their finite compositions preserve $\mathcal{D}$.

Proof. It suffices to use one matter vector and one ghost monomial. For the matter term, indices $n \leq -2$ require the occupied creator b_n . Only finitely many such indices occur. The restrictedness of $\mathcal{M}$ bounds the remaining positive indices.

For a normally ordered ghost monomial, every annihilator must remove an occupied creator. Its indices therefore lie in a fixed finite set. All creator indices have an upper bound: -2 for b and 1 for c . The sum of the indices is fixed in each displayed operator. Once the annihilator indices are fixed, this sum and the upper bounds give lower bounds for every creator index. There are at most three factors. Hence only finitely many terms act nontrivially. The argument includes terms containing only creators. It also applies to ν_r and K . Each output is a finite linear combination of algebraic states, so compositions are defined on the same domain. $\square$

For trivial matter, all $L_n^{\mathcal{M}}$ vanish and $C = 0$. The first nonzero anomalous mode is visible without an infinite summation:

$$\begin{aligned} R\Omega &= 0, & Rb_{-2}\Omega &= -2b_{-2}c_0\Omega - b_{-3}c_1\Omega, \\ R(b_{-2}c_0\Omega) &= 4c_{-2}\Omega + b_{-3}c_0c_1\Omega + 2b_{-2}c_{-1}c_1\Omega, \\ R(b_{-3}c_1\Omega) &= 5c_{-2}\Omega - 2b_{-3}c_0c_1\Omega - 4b_{-2}c_{-1}c_1\Omega. \end{aligned}$$

Consequently,

$$R^2b_{-2}\Omega = -13c_{-2}\Omega. \quad (1.6)$$

The cancellation of the cubic output leaves a degree-two operator acting on a degree-minus-one input. The calculation below determines every mode of this operator.

1.2 THE GHOST CENTRAL TERM

LEMMA 1.2.1 (*Ghost commutants*). An operator on $\mathcal{M} \otimes \mathcal{F}$ that supercommutes with every b_n, c_n is zero if its ghost degree is nonzero. An even operator on $\mathcal{F}$ commuting with every b_n, c_n is a scalar.

Proof. The common kernel of the vacuum annihilators on $\mathcal{M} \otimes \mathcal{F}$ is $\mathcal{M} \otimes \mathbb{C}\Omega$. Indeed, these annihilators are all the exterior derivatives of the creator variables. A finite exterior polynomial killed by all these derivatives is constant. A supercommuting operator sends this common kernel into itself. The kernel has ghost degree zero, so an operator of nonzero ghost degree vanishes there. Supercommutation with the creators makes it vanish on every state. For an even operator on $\mathcal{F}$, its value on Ω is a scalar multiple of Ω . Commutation with the creators gives the same scalar on all monomials. $\square$

PROPOSITION 1.2.2 (*The ghost Virasoro algebra*). On $\mathcal{F}$,

$$[L_m^{\text{gh}}, b_n] = (m - n)b_{m+n}, \quad [L_m^{\text{gh}}, c_n] = (-2m - n)c_{m+n}, \quad (1.7)$$

$$L_0^{\text{gh}} = E, \quad L_m^{\text{gh}}\Omega = 0 \quad (m \geq -1), \quad (1.8)$$

$$[L_m^{\text{gh}}, L_n^{\text{gh}}] = (m - n)L_{m+n}^{\text{gh}} - \frac{26}{12}(m^3 - m)\delta_{m+n,0} \mathbf{1}. \quad (1.9)$$

Proof. The elementary contractions are

$$[: b_r c_s :, b_n] = \delta_{s+n,0} b_r, \quad [: b_r c_s :, c_n] = -\delta_{r+n,0} c_s.$$

They give (1.7). A term acting on Ω needs both factors to be creators. For $m \geq 0$ this is impossible. For $m = -1$ the only possible term has coefficient zero. The commutators of L_0^{gh} with the creators equal those of E . Both operators kill Ω , which proves (1.8).

Let $A_{m,n} = [L_m^{\text{gh}}, L_n^{\text{gh}}] - (m - n)L_{m+n}^{\text{gh}}$. Substitution of (1.7) into the Jacobi identity shows that $A_{m,n}$ commutes with every ghost generator. For example, its commutator with b_r has coefficient

$$(n - r)(m - n - r) - (m - r)(n - m - r) - (m - n)(m + n - r) = 0.$$

For c_r , the coefficient is

$$(-2n - r)(-2m - n - r) - (-2m - r)(-2n - m - r) - (m - n)(-2m - 2n - r) = 0.$$

Lemma 1.2.1 makes $A_{m,n}$ a scalar. Its energy is $-m - n$, so this scalar vanishes unless $m + n = 0$.

For $m \geq 2$, the complete vacuum expression is

$$L_{-m}^{\text{gh}}\Omega = \sum_{s=2-m}^{-1} (s - m)b_{-m-s}c_s\Omega.$$

Commuting L_m^{gh} through the first factor gives $(2m + s)b_{-s}$. This contracts with c_s to give 1. Its commutator with the second factor annihilates Ω . Therefore

$$\begin{aligned} A_{m,-m}\Omega &= \sum_{s=2-m}^{-1} (s - m)(2m + s)\Omega \\ &= \left(\sum_{s=2-m}^{-1} s^2 + m \sum_{s=2-m}^{-1} s - 2m^2 \cdot m \right) \Omega \\ &= -\frac{13}{6}(m^3 - m)\Omega. \end{aligned}$$

Here the interval has m integers, $\sum s = m(3 - m)/2$, and $\sum s^2 = m(2m^2 - 9m + 13)/6$. The cases $m = 0, 1$ give zero by (1.8). Antisymmetry gives negative m . This proves (1.9). $\square$

Set $\mathcal{L}_n = L_n^M + L_n^{\text{gh}}$ on $\mathcal{D}$. The two summands commute. Thus (1.1) and (1.9) give

$$[\mathcal{L}_m, \mathcal{L}_n] = (m - n)\mathcal{L}_{m+n} + A(m)\delta_{m+n,0}, \quad A(m) = \frac{C - 26}{12}(m^3 - m). \quad (1.10)$$

This central term determines the square of Q on the full ghost module.

1.3 THE SQUARE AS AN OPERATOR IDENTITY

LEMMA 1.3.1 (*Contractions of the charge*). The operator (1.5) satisfies

$$\{Q, b_n\} = \mathcal{L}_n, \quad \{Q, c_n\} = \nu_n. \quad (1.11)$$

Moreover,

$$[\mathcal{L}_m, \nu_n] = (-2m - n)\nu_{m+n}, \quad [Q, \nu_n] = 0. \quad (1.12)$$

Proof. Contracting b_n with the two c factors in R gives

$$\{R, b_n\} = \sum_{m \in \mathbb{Z}} (m - n) : c_{-m} b_{m+n} := \sum_{s \in \mathbb{Z}} (n + s) : b_{n-s} c_s := L_n^{\text{gh}}.$$

Contracting c_n with the b factor gives

$$\{R, c_n\} = -\frac{1}{2} \sum_{r+s=-n} (r - s) c_{-r} c_{-s} = \nu_n.$$

The matter term contributes L_n^M to the first identity and zero to the second. All contractions retain the same normal ordering of the uncontracted factors.

The commutator of $\mathcal{L}_m$ with ν_n uses (1.7). After reindexing, the coefficient of $c_u c_v$, $u + v = m + n$, is

$$\frac{1}{2} ((u - v - m)(-m - u) + (u - v + m)(-m - v)) = -\frac{1}{2} (2m + n)(u - v).$$

This proves the first part of (1.12).

For the second part, extend $\delta(c_n) = \nu_n$ as an odd derivation on products of c modes. Their mutual anticommutators are zero. The coefficient of each alternating cubic monomial in $\delta^2(c_n)$ is a multiple of

$$(a - b)(a + b - d) + (b - d)(b + d - a) + (d - a)(d + a - b) = 0.$$

Thus $\delta^2(c_n) = 0$. The second identity in (1.11) identifies $[Q, \nu_n]$ with this expression. Lemma 1.1.1 permits these expansions on each algebraic input. $\square$

THEOREM 1.3.2 (*The Virasoro square*). For every restricted matter module $\mathcal{M}$ as above,

$$Q^2 = \frac{C - 26}{12} \sum_{n > 0} (n^3 - n) c_{-n} c_n \quad \text{on } \mathcal{M} \otimes \mathcal{F}. \quad (1.13)$$

In addition,

$$[Q, \mathcal{L}_n] = -A(n)c_n, \quad [Q, \mathcal{L}_0] = 0. \quad (1.14)$$

If $\mathcal{M} \neq 0$, the operator Q is square-zero on the full state space if and only if $C = 26$.

Proof. Put $D_m = [Q, \mathcal{L}_m]$. The graded Jacobi identity, (1.11), and (1.10) give

$$\begin{aligned} \{D_m, b_n\} &= [Q, [\mathcal{L}_m, b_n]]_s - [\mathcal{L}_m, \{Q, b_n\}] \\ &= (m - n)\mathcal{L}_{m+n} - [\mathcal{L}_m, \mathcal{L}_n] = -A(m)\delta_{m+n,0}, \\ \{D_m, c_n\} &= (-2m - n)\nu_{m+n} - [\mathcal{L}_m, \nu_n] = 0. \end{aligned}$$

Hence $D_m + A(m)c_m$ supercommutes with all ghosts. It has ghost degree one, so Lemma 1.2.1 makes it zero. This proves (1.14).

Let K be the right side of (1.13). A direct Clifford contraction gives

$$[K, b_n] = -A(n)c_n, \quad [K, c_n] = 0.$$

For instance, $[c_{-m}c_m, b_n] = \delta_{m+n,0}c_{-m} - \delta_{n-m,0}c_m$. The oddness of A gives the formula for both signs of n . Also,

$$[Q^2, b_n] = [Q, \{Q, b_n\}] = [Q, \mathcal{L}_n], \quad [Q^2, c_n] = [Q, v_n] = 0.$$

Thus $Q^2 - K$ commutes with all ghosts and has ghost degree two. Lemma 1.2.1 proves that it is zero.

When $C = 26$, the displayed square vanishes. For nonzero $v \in M$, the converse follows from

$$Q^2(v \otimes b_{-2}\Omega) = \frac{C-26}{2} v \otimes c_{-2}\Omega.$$

The output is nonzero when $C \neq 26$. □

1.4 FIELDS, ZERO MODES, AND THE INTERCEPT

Fix a formal coordinate z and define

$$b(z) = \sum_n b_n z^{-n-2}, \quad c(z) = \sum_n c_n z^{1-n}, \quad T^M(z) = \sum_n L_n^M z^{-n-2}.$$

The residue extracts the coefficient of $z^{-1}dz$. For convergent Laurent series it equals the positively oriented integral divided by $2\pi i$. All uses here are formal. The two ghost contractions are the expansion of $(z-w)^{-1}$ in nonnegative powers of w/z . For example,

$$\sum_{n \geq -1} z^{-n-2} w^{n+1} = \frac{1}{z-w}.$$

The ghost stress tensor is

$$T^{\text{gh}}(z) = -2 : b(z) \partial c(z) : - : \partial b(z) c(z) : = \sum_m L_m^{\text{gh}} z^{-m-2}.$$

Indeed, the coefficient of $b_r c_s$ is $r + 2s$. When $r + s = m$, this is $m + s$, as in (1.3). Equation (1.7) specifies the conformal weights 2 and -1.

Multiplying the fields and taking the residue gives exactly

$$Q = \text{Res}_{z=0} (c(z) T^M(z) + : b(z) c(z) \partial c(z) :) dz. \quad (1.15)$$

For the matter term, the residue condition is $m + n = 0$. For the ghost term it is $r + s + t = 0$. Antisymmetrizing the two c factors changes the coefficient $1 - t$ to $(s - t)/2$, which gives (1.4). If $\mathcal{N}$ means full normal ordering of the elementary ghost factors, then

$$\frac{1}{2} \mathcal{N}(c T^{\text{gh}}) = : bc \partial c :.$$

Thus (1.15) also gives the residue of $c T^M + \frac{1}{2} \mathcal{N}(c T^{\text{gh}})$. Adding $\frac{3}{2} \partial^2 c$ to this current leaves its residue unchanged. This specifies a zero mode, rather than a sum of all current modes.

PROPOSITION I.4.1 (*Intercept and vacuum conventions*). For $t \in \mathbb{C}$, set $Q_t = Q + tc_0$. Then

$$\{Q_t, b_n\} = \mathcal{L}_n + t\delta_{n,0}, \quad [Q_t, \mathcal{L}_0] = 0, \quad (I.16)$$

$$Q_t^2 = \sum_{n>0} \left(\frac{C-26}{12} (n^3 - n) - 2tn \right) c_{-n}c_n. \quad (I.17)$$

For $M \neq 0$, absolute nilpotence holds exactly when $C = 26$ and $t = 0$.

Let $\downarrow = c_1\Omega$ and normal-order relative to this vacuum. Its annihilators are b_n for $n \geq 0$ and c_n for $n \geq 1$. Use a subscript $\downarrow$ for this ordering. Then

$$L_{n,\downarrow}^{\text{gh}} = L_n^{\text{gh}} + \delta_{n,0}, \quad R_{\downarrow} = R + c_0, \quad \sum_n c_{-n}L_n^M + R_{\downarrow} - ac_0 = Q + (1-a)c_0. \quad (I.18)$$

In this convention the nilpotent intercept is $a = 1$. The underlying Clifford module is the same. The vacuum $\downarrow$ has conformal energy -1 and ghost degree one in the Ω convention.

Proof. The first line follows from (I.11) and $[\mathcal{L}_0, c_0] = 0$. Since $c_0^2 = 0$,

$$Q_t^2 = Q^2 + t\{Q, c_0\}, \quad \{Q, c_0\} = \nu_0 = -2 \sum_{n>0} nc_{-n}c_n.$$

This proves (I.17). On $v \otimes \Omega$, its value is $-2tv \otimes c_{-1}c_1\Omega$. Thus absolute nilpotence requires $t = 0$. Theorem I.3.2 then requires $C = 26$.

The change of vacuum reverses only the creator/annihilator pair (c_1, b_{-1}) . Its normal products satisfy

$$: b_{-1}c_1 :_{\downarrow} - : b_{-1}c_1 := b_{-1}c_1 + c_1b_{-1} = 1.$$

The coefficient of this pair in L_0^{gh} is 1, and it occurs in no other L_n^{gh} . This gives the first identity in (I.18). The same contraction proof as in Lemma I.3.1 gives

$$\{R_{\downarrow} - R, b_n\} = \delta_{n,0}, \quad \{R_{\downarrow} - R, c_n\} = 0.$$

The difference $R_{\downarrow} - R - c_0$ vanishes by Lemma I.2.1. The last identity in (I.18) follows. The energy and degree of $\downarrow$ follow from those of c_1 . $\square$

PROPOSITION I.4.2 (*The nonnegative ghost polarization*). Set $c^m = c_{-m}$, so that $\{b_m, c^n\} = \delta_m^n$. Let $\mathcal{F}_{\text{pol}}$ have vacuum Ω_{pol} killed by c^m for $m \leq 0$ and by b_m for $m > 0$. Its creators are c^m for $m > 0$ and b_m for $m \leq 0$. Give its vacuum degree and energy zero, and assign these creators degrees 1, -1 and energies $m, -m$ respectively. There is a Clifford-module isomorphism

$$I : \mathcal{F}_{\text{pol}} \longrightarrow \mathcal{F}, \quad I(x\Omega_{\text{pol}}) = x c_0 c_1 \Omega,$$

where x is a product of the indicated creators. It raises ghost degree by 2 and lowers energy by 1. Let R_{pol} denote the cubic expression (I.4) with this ordering. Under $I_M \otimes I$,

$$\sum_m c^m L_m^M + R_{\text{pol}} - ac^0 \longmapsto Q + (1-a)c_0.$$

Thus its critical charge term is $-c^0$, with both degree and energy shifts specified by I .

Proof. The vector $\Xi = c_0 c_1 \Omega$ obeys the required annihilator conditions. For $m \leq 0$, $c_{-m} \Xi = 0$: c_0 and c_1 square to zero, and all c_n with $n \geq 2$ annihilate Ξ . For $m > 0$, $b_m \Xi = 0$ by the Clifford relations and (1.2). Hence the displayed map intertwines the Clifford actions. Away from the pairs (b_0, c_0) and (b_{-1}, c_1) it identifies the same exterior creators. On each of these two pairs, exterior multiplication and exterior differentiation exchange the empty and occupied states, with nonzero signs. Their four-state tensor product is invertible. This proves that I is an isomorphism. Since Ξ has degree 2 and energy -1 , the grading statements follow.

Relative to Ω , this ordering reverses both specified pairs. The pair (b_{-1}, c_1) contributes 1 to L_0^{gh} , as in Proposition 1.4.1. The pair (b_0, c_0) has coefficient zero there. Consequently the transported ghost stress modes are $L_n^{\text{gh}} + \delta_{n,0}$. The contraction identities and Lemma 1.2.1 then give $I R_{\text{pol}} I^{-1} = R + c_0$. The matter term commutes with I , which proves the charge formula. $\square$

1.5 ABSOLUTE AND RELATIVE COMPLEXES

Here absolute and relative refer to the b_0 constraint. The central character of the Virasoro module is fixed, and the ghost variables are indexed by the modes L_n . There is no ghost for the central generator in this carrier.

Set

$$\mathcal{D}_t^{\text{rel}} = \ker(b_0 : \mathcal{D} \rightarrow \mathcal{D}) \cap \ker(\mathcal{L}_0 + t : \mathcal{D} \rightarrow \mathcal{D}).$$

It inherits the ghost grading of $\mathcal{D}$. The first identity in (1.16) and the commutation with $\mathcal{L}_0$ show that Q_t preserves this subspace. On it, the square is the restriction of (1.17).

COROLLARY 1.5.1 (*The critical complexes*). At $C = 26$ and $t = 0$, the pairs

$$C_{\text{abs}}^q(M) = M \otimes \mathcal{F}^q, \quad C_{\text{rel}}^q(M) = \mathcal{D}_0^{\text{rel}} \cap (M \otimes \mathcal{F}^q)$$

with differential Q are cochain complexes of complex vector spaces. The inclusion $C_{\text{rel}}^\bullet(M) \rightarrow C_{\text{abs}}^\bullet(M)$ is a chain map. A homomorphism of restricted Virasoro modules induces chain maps by tensoring with $1_{\mathcal{F}}$. If $\mathcal{L}_0$ acts semisimply, every nonzero weight λ subcomplex of the absolute complex is contractible by $\lambda^{-1} b_0$.

Proof. The domain lemma defines the differential in every degree. Theorem 1.3.2 gives its square, and (1.16) gives stability of the relative subspace. A module homomorphism commutes termwise with (1.5), hence with the finite sums on each state. On the weight- λ subspace, $Q(\lambda^{-1} b_0) + (\lambda^{-1} b_0)Q = 1$ by (1.11). $\square$

Relative nilpotence depends on the actual relative carrier. The full-state converse in Theorem 1.3.2 cannot be restricted without checking its detecting vectors.

PROPOSITION 1.5.2 (*Nonnegative matter weights*). Suppose $M = \bigoplus_{j \geq 0} M_j$, with $L_0^M|_{M_j} = j$ and $L_n^M M_j \subseteq M_{j-n}$. At $t = 0$, for every central charge C , the relative carrier is

$$\mathcal{D}_0^{\text{rel}} = (M_0 \otimes \mathbb{C}\Omega) \oplus (M_1 \otimes \mathbb{C}c_1\Omega) \oplus (M_0 \otimes \mathbb{C}c_{-1}c_1\Omega). \quad (1.19)$$

Its differential is the three-term complex

$$M_0 \xrightarrow{L_{-1}^M} M_1 \xrightarrow{L_1^M} M_0 \quad (1.20)$$

in ghost degrees 0, 1, 2. In particular, $Q^2 = 0$ on this relative carrier for every C .

Proof. The only negative ghost energy is that of c_1 , and it can occur at most once. Thus $\mathcal{F}$ has minimum energy -1 . Every b creator has energy at least 2, so no b creator occurs at ghost energy 0 or -1 . The complete bases at these two energies are

$$\mathcal{F}_{-1} = \langle c_1 \Omega, c_0 c_1 \Omega \rangle, \quad \mathcal{F}_0 = \langle \Omega, c_0 \Omega, c_{-1} c_1 \Omega, c_{-1} c_0 c_1 \Omega \rangle.$$

The condition $b_0 = 0$ removes precisely the terms containing c_0 . The total weight-zero condition therefore gives (1.19).

The matter term and ghost term of (1.5) give

$$\begin{aligned} Q(v \otimes \Omega) &= L_{-1}^M v \otimes c_1 \Omega & (v \in M_0), \\ Q(w \otimes c_1 \Omega) &= L_1^M w \otimes c_{-1} c_1 \Omega & (w \in M_1), \\ Q(v \otimes c_{-1} c_1 \Omega) &= 0 & (v \in M_0). \end{aligned}$$

In the middle line, $c_0 L_0^M w \otimes c_1 \Omega$ cancels $w \otimes R c_1 \Omega = -w \otimes c_0 c_1 \Omega$. For the last line, $R(c_{-1} c_1 \Omega) = 0$ by the two contractions in Lemma 1.3.1. Finally, $L_1^M L_{-1}^M v = 2L_0^M v + L_{-1}^M L_1^M v = 0$ on M_0 . This proves (1.20) and its nilpotence. $\square$

For example, the universal Virasoro vacuum module of central charge C is induced from a vector $\mathbf{1}$ killed by L_n for $n \geq -1$. The central generator acts by C . The ordered monomials in L_{-n} , $n \geq 2$, form its basis by the Poincaré–Birkhoff–Witt theorem. Hence $M_0 = \mathbb{C}\mathbf{1}$ and $M_1 = 0$. Its relative complex has basis $\mathbf{1} \otimes \Omega$, $\mathbf{1} \otimes c_{-1} c_1 \Omega$ and zero differential for every C . Trivial matter gives the same relative complex at $C = 0$. Its absolute square is nonzero by (1.6). At the other boundary, suppose a restricted module contains $v \neq 0$ with $L_0^M v = -2v$. Then $v \otimes b_{-2} \Omega$ lies in $\mathcal{D}_0^{\text{rel}}$ and

$$Q^2(v \otimes b_{-2} \Omega) = \frac{C - 26}{2} v \otimes c_{-2} \Omega.$$

This relative square is nonzero when $C \neq 26$. Thus the critical central charge guarantees relative nilpotence for every restricted module, while particular relative carriers can have further vanishing.

1.6 CONFORMAL VERTEX ALGEBRA STATES

Let V be an ordinary conformal vertex algebra over $\mathbb{C}$ with conformal vector ω . Write $Y(\omega, z) = \sum_n L_n^V z^{-n-2}$ and let C be its central charge. The lower truncation axiom makes V a restricted matter module. Consequently (1.5) defines the degree-one operator on the full algebraic state carrier $V \otimes \mathcal{F}$. Equation (1.15) identifies it with the zero mode of the specified normal-ordered current. At $C = 26$, Corollary 1.5.1 supplies both the absolute complex and its b_0 -relative subcomplex. When V has nonnegative integral conformal weights, Proposition 1.5.2 computes the relative complex directly.

The ghost degree here is the cohomological degree. Conformal weight is the eigenvalue of $L_0^V + L_0^{\text{gh}}$ and is independent of ghost degree. The residue uses the fixed formal coordinate z . A comparison with a complex of collision-state sheaves requires maps in that sheaf category, with compatible restrictions and differentials. The state-space construction above supplies its Virasoro differential and its square.

1.6.1 Augmentations and the relative coordinate carrier

The critical Virasoro charge acts on an algebraic state space. A comparison with chiral chains also needs a target and compatible coordinate transitions. Two finite identities obstruct the augmentation-kernel target and the standard restricted transitions.

PROPOSITION 1.6.1 (*The augmentation-kernel obstruction*). Let $V \neq 0$ be a unital vertex algebra over $\mathbb{C}$, and let $A = V \otimes bc$ use the ghost system above. There is no unital vertex superalgebra homomorphism $A \rightarrow \mathbb{C}$ with trivial target vertex products. Consequently the reduced chiral construction whose coefficient sheaf is a chiral augmentation kernel cannot be instantiated on A with its original collision products.

Proof. Write $\Omega_A = \mathbf{1}_V \otimes \Omega$, $b = \mathbf{1}_V \otimes b_{-2}\Omega$, and $c = \mathbf{1}_V \otimes c_1\Omega$ in A . With the vertex convention $Y(a, z) = \sum_r a_{(r)} z^{-r-1}$,

$$b_{(0)}c = \mathbf{1}_V \otimes b_{-1}c_1\Omega = \Omega_A.$$

Every nonnegative product vanishes in the trivial vertex algebra. A unital homomorphism ε would therefore give $1 = \varepsilon(b_{(0)}c) = \varepsilon(b)_{(0)}\varepsilon(c) = 0$. For $V = \text{Vir}_{26}$, set $T = (L_{-2}^V \mathbf{1}_V) \otimes \Omega$. There is also the independent matter identity

$$T_{(3)}T = (L_2^V L_{-2}^V \mathbf{1}_V) \otimes \Omega = 13\Omega_A.$$

On a coordinate disk, let $t = z_1 - z_2$ and normalize $\text{Res}(t^{-1}dt) = 1$. The full collision maps include

$$\text{Res}_{t=0} Y(b, t)c dt = \Omega_A, \quad \text{Res}_{t=0} t^3 Y(T, t)T dt = 13\Omega_A.$$

A chiral augmentation to the unit chiral algebra must preserve these residues, which gives the same contradiction on that disk. It therefore cannot exist globally. The forms $dt = t d \log t$ and $t^3 dt = t^4 d \log t$ have regular logarithmic coefficients. Restricting to such coefficients does not remove the obstruction.

The linear vacuum projection $\pi : A \rightarrow \mathbb{C}$ has $b, c \in \ker \pi$ but $b_{(0)}c \notin \ker \pi$. Its kernel is not closed under the original collision maps. The quotient by the vacuum also fails to inherit all vertex products, since $(\Omega_A)_{(-1)}b = b$ whereas $0_{(-1)}b = 0$. Thus neither linear operation supplies the missing chiral augmentation. $\square$

PROPOSITION 1.6.2 (*Relative states, absolute states, and coordinate changes*). Take $V = \text{Vir}_{26}$ and the zero-intercept charge of Theorem 1.3.2. In this proposition, Ω denotes the tensor-product vacuum $\mathbf{1}_V \otimes \Omega_{\text{gh}}$. Put $H = L_o^V + L_o^{\text{gh}}$ and

$$e = \Omega, \quad u = c_o\Omega, \quad v = c_{-1}c_1\Omega, \quad w = c_{-1}c_o c_1\Omega.$$

The weight-zero absolute complex has these four basis states and differential

$$Qe = 0, \quad Qu = -2v, \quad Qv = Qw = 0.$$

The relative complex is $\mathbb{C}e \oplus \mathbb{C}v$ with zero differential. The absolute cohomology has one class in degrees zero and three, and no others. The inclusion of relative states into absolute states kills $[v]$ and is not a quasi-isomorphism.

Fix the convention that the coordinate flow $z \mapsto z/(1 + sz)$ acts by $g_s = \exp(s\mathcal{L}_1)$, where $\mathcal{L}_1 = L_1^V + L_1^{\text{gh}}$. Then

$$g_s v = v - s c_o c_1 \Omega, \quad b_o g_s v = -s c_1 \Omega \neq 0 \quad (s \neq 0). \quad (1.21)$$

The fixed relative fiber therefore does not define a subbundle through restriction of these state-bundle transitions.

Proof. The universal vacuum module has $V_0 = \mathbb{C}\mathbf{1}$ and $V_1 = 0$. The energy argument of Proposition 1.5.2 gives exactly the four displayed absolute states and the two relative states. The charge contraction gives $Qu = -2v$. Nilpotence gives $Qv = 0$, and weight zero has no degree-four state, so $Qw = 0$. At every nonzero total weight λ , the homotopy b_0/λ contracts the absolute complex by Corollary 1.5.1. On the four-state complex, the homotopy $h(v) = -u/2$, zero on e, u, w , contracts it onto $\mathbb{C}e \oplus \mathbb{C}w$. All state sums are algebraic, so these homotopies define maps on the full carrier. In particular, $v = Q(-u/2)$ in the absolute complex.

The ghost transformation law (1.7) gives

$$\mathcal{L}_1 v = -c_0 c_1 \Omega, \quad \mathcal{L}_1 (c_0 c_1 \Omega) = 0.$$

The exponential on v consequently terminates after its linear term. Since $b_0 c_0 c_1 \Omega = c_1 \Omega$, this proves (1.21). Also $H g_5 v = s c_0 c_1 \Omega$, so the same transition fails the weight-zero condition. A restricted transition must preserve the model fiber, which fails here. $\square$

For a fixed coordinate, the relative vector-space complex remains defined. A sheaf comparison requires a separately constructed relative sheaf complex or a family of transported constraints with compatible transition maps. For source and target sheaf complexes $\mathcal{C}, \mathcal{T}$, local maps must satisfy

$$d_{\mathcal{T}} \Phi_U = \Phi_U d_{\mathcal{C}}, \quad \text{res}_{UW}^{\mathcal{T}} \Phi_U = \Phi_W \text{res}_{UW}^{\mathcal{C}} \quad (W \subset U),$$

and $g_{ij}^{\mathcal{T}} \Phi_j = \Phi_i g_{ij}^{\mathcal{C}}$ in coordinate trivializations. Ordinary or derived global sections require these complexes and their transitions first. The restricted transitions in (1.21) do not supply them.

A different unital chiral-chain construction may avoid an augmentation kernel. Its coefficient complexes, vacuum insertions, all collision maps, differential, and section functor must be constructed before it can serve as a comparison target. For the relative vacuum complex, a degree-preserving quasi-isomorphism requires target cohomology precisely in degrees zero and two. Proposition 1.6.2 excludes the absolute state complex as a replacement target. These assertions concern the specified augmentation-kernel construction and state transitions; they do not exclude every chiral-chain construction.

1.7 HAMILTONIAN REDUCTION AND COMPARISON OF COMPLEXES

The constraint $e = 1$ for $\mathfrak{sl}_2$ gives a reduction whose maps can be written on every polynomial. It separates Hamiltonian reduction from the Virasoro charge and from chiral collision operations. All complexes in the polynomial calculation are algebraic cochain complexes over $\mathbb{C}$. Their products use the Koszul sign $(-1)^{|x||y|}$.

1.7.1 The polynomial reduction

Let e, h, f be the linear coordinate functions on $\mathfrak{sl}_2^*$, with Poisson brackets

$$\{h, e\} = 2e, \quad \{h, f\} = -2f, \quad \{e, f\} = h.$$

Introduce odd generators b, c of cohomological degrees $-1, 1$. The even generators have degree zero. On the graded commutative algebra

$$A = \mathbb{C}[e, h, f] \otimes \Lambda(b, c)$$

define a degree-one derivation D by

$$De = 0, \quad Dh = -2ec, \quad Df = hc, \quad Db = e - 1, \quad Dc = 0.$$

Thus the Koszul differential imposes $e = 1$, and the other term is the Chevalley–Eilenberg differential for the Hamiltonian vector field $\{e, -\}$. The two terms anticommute. Direct substitution on the five generators gives $D^2 = 0$. Let $R = \mathbb{C}[u]$ have zero differential and augmentation $u \mapsto 0$. Give A the augmentation

$$\varepsilon(e) = 1, \quad \varepsilon(h) = \varepsilon(f) = \varepsilon(b) = \varepsilon(c) = 0.$$

These choices concern the displayed differential graded commutative algebra. They do not give an augmentation of an affine or ghost vertex algebra.

THEOREM 1.7.1 (*Polynomial reduction and its bar comparison*). The maps of augmented differential graded algebras

$$i : R \longrightarrow A, \quad i(u) = ef + \frac{1}{4}h^2, \quad (1.22)$$

$$p : A \longrightarrow R, \quad (p(e), p(h), p(f), p(b), p(c)) = (1, 0, u, 0, 0) \quad (1.23)$$

satisfy $pi = 1$ and are quasi-isomorphisms. In particular,

$$H^j(A, D) = \begin{cases} \mathbb{C}[u], & j = 0, \\ 0, & j \neq 0. \end{cases}$$

The induced maps $B(i)$ and $B(p)$ on the direct-sum associative reduced bar constructions are quasi-isomorphisms, and $B(p)B(i) = 1$.

Proof. The element $K = ef + h^2/4$ satisfies $DK = ehc - ehc = 0$. The projection kills the image under D of every generator, including $Db = e - 1$. This proves the chain identities and $pi = 1$.

Resolve the differential as

$$D = (e - 1)\partial_b + c(-2e\partial_h + h\partial_f).$$

Filter by the number of c factors. The filtration has two steps, so its spectral sequence has no convergence issue. Its first cohomology calculation is the two-term Koszul complex for $e - 1$. Multiplication by $e - 1$ is injective on the polynomial ring. The resulting complex is

$$\mathbb{C}[h, f] \xrightarrow{-2\partial_h + h\partial_f} c\mathbb{C}[h, f].$$

Set $u = f + h^2/4$. This is a polynomial change of coordinates from (h, f) to (h, u) . In these coordinates the displayed derivative is $-2\partial_h$. Its kernel is $\mathbb{C}[u]$, and it is surjective in characteristic zero. For example, the primitive of $ch^a u^j$ is $-h^{a+1}u^j/(2a+2)$. Thus cohomology is concentrated in degree zero, and p induces its displayed identification. The identity $pi = 1$ proves the same assertion for i .

Here are the bar conventions and the induced maps. For an augmented differential graded algebra E , put $\bar{E} = \ker \varepsilon$ and

$$B(E) = \bigoplus_{n \geq 0} (\bar{E}[1])^{\otimes n}.$$

Write sa for a considered in $\bar{E}[1]$, so $|sa| = |a| - 1$. The bar coderivation has components

$$b_1(sa) = -s(d_E a), \quad b_2(sa \otimes sb) = (-1)^{|a|} s(ab).$$

Extend each component to consecutive positions with the Koszul sign for a degree-one operator. The Leibniz rule and associativity give $b^2 = 0$. For an augmented chain algebra map q , set

$$B(q)(sa_1 \otimes \cdots \otimes sa_n) = sq(a_1) \otimes \cdots \otimes sq(a_n).$$

The identities $qd_E = d_F q$ and $q(ab) = q(a)q(b)$ prove that $B(q)$ commutes with b_1 and b_2 separately.

Filter the cone of $B(p)$ by tensor length. Its length- n quotient is the cone of $(\tilde{p}[1])^{\otimes n}$ with the internal differentials. The splitting $E = \mathbb{C} \oplus \tilde{E}$ shows that $\tilde{p}$ is a quasi-isomorphism. Tensor products of complexes of vector spaces preserve quasi-isomorphisms. One proof splits every acyclic complex into two-term contractible summands and tensors its contraction with the identity. Consequently each length quotient of the cone is acyclic. A cone cycle has finite maximal length. Solve its boundary equation in that highest quotient and subtract the resulting boundary. This lowers its maximal length. Induction ends at length zero and proves that the whole cone is acyclic. The same argument applies to $B(i)$, and functoriality gives $B(p)B(i) = 1$. $\square$

The bar argument uses finite tensor words and polynomial primitives. A product completion, a vertex algebra, and a chiral sheaf each require a separate comparison. Even on $\mathcal{A}$, the projection p does not preserve the original Poisson brackets:

$$p\{h, f\} = -2u, \quad \{p(h), p(f)\} = 0.$$

Hence this chain algebra map cannot transport a Poisson or Batalin–Vilkovisky operation merely by forgetting its source.

1.7.2 Reduction of a specified double complex

Two odd differentials require their anticommutator to vanish on their common domain. Their separate nilpotence does not imply this equation. For example, on $\Lambda(x, y)$, exterior multiplication $d_1 = x \wedge -$ and left differentiation $d_2 = \partial_x$ are odd and square zero, but $d_1 d_2 + d_2 d_1 = 1$.

THEOREM 1.7.2 (*A comparison under a second differential*). Let $X^{p,q}, Y^{p,q}$ be bigraded complex vector spaces, zero unless $0 \leq p \leq N$. Let d have bidegree $(1, 0)$ and δ have bidegree $(0, 1)$, with

$$d^2 = \delta^2 = 0, \quad d\delta + \delta d = 0.$$

Suppose $F : X \rightarrow Y$ has bidegree $(0, 0)$, commutes with both differentials, and induces a quasi-isomorphism on every column $(X^{p,*}, \delta)$. Then

$$\text{Tot}(F) : \left(\bigoplus_{p+q=*} X^{p,q}, d + \delta \right) \longrightarrow \left(\bigoplus_{p+q=*} Y^{p,q}, d + \delta \right)$$

is a quasi-isomorphism. The formula is $\text{Tot}(F)((x^{p,q})) = (F(x^{p,q}))$.

Proof. Filter the total complexes decreasingly by the first degree p . This filtration is finite, and the differential on each quotient is δ . The same filtration on the mapping cone has acyclic quotients by hypothesis. The short exact sequence of two successive filtration stages gives a long exact sequence in cohomology. Induction through the finite filtration proves that the cone is acyclic. The two commutation identities give the total chain identity before any cohomology is taken. $\square$

For quantum Drinfeld–Sokolov reduction, the input category in Arakawa’s vanishing theorem is KL_k . It consists of the specified graded affine modules with locally finite action of the finite-dimensional Lie algebra. The theorem gives $H_f^j(M) = 0$ for $j \neq 0$ and exactness of H_f^0 on this category [1, §6, p. 585; Theorem 7.1(i), p. 588]. Applying it to a collision complex requires an affine action on each term, membership in this category, and equivariance of all collision maps. To use the displayed totalization, one also needs its boundedness hypothesis, or a proved substitute for convergence. The nonlinear $\mathcal{W}$ -algebra and a conformal grading alone do not specify a second semi-infinite differential.

The polynomial maps (1.22)–(1.23) construct the associative comparison for the classical $\mathfrak{sl}_2$ constraint. A chiral comparison additionally needs maps from its collision sheaves, including the vacuum terms of every operator product. The augmentation obstruction in Proposition 1.6.1 excludes the augmentation-kernel construction for the matter–ghost vertex algebra.

1.7.3 The Virasoro target at a reduced level

COROLLARY 1.7.3 (*Nilpotence on the Virasoro vacuum carrier*). Let $k \in \mathbb{C} \setminus \{-2\}$ and let $M_{c(k)}$ be the universal Virasoro vacuum module with

$$c(k) = 1 - \frac{6(k+1)^2}{k+2}.$$

Use the ghost module and charge Q of Theorem 1.3.2 on $M_{c(k)} \otimes \mathcal{F}$. This charge squares to zero on the full algebraic carrier if and only if

$$k = -\frac{8}{3} \quad \text{or} \quad k = -\frac{7}{2}.$$

At either level, its absolute vacuum cohomology has one-dimensional components in degrees 0, 3. Its relative vacuum cohomology has one-dimensional components in degrees 0, 2. All other components vanish.

Proof. Theorem 1.3.2 gives

$$Q^2 = \frac{c(k) - 26}{12} \sum_{n>0} (n^3 - n) c_{-n} c_n.$$

On $\mathbf{1} \otimes b_{-2}\Omega$ the output is $(c(k) - 26)\mathbf{1} \otimes c_{-2}\Omega/2$, a nonzero vector when $c(k) \neq 26$. The equation $c(k) = 26$ is $6k^2 + 37k + 56 = (3k+8)(2k+7) = 0$. At these levels the matter module is the universal vacuum module of central charge 26. The four-state calculation and the nonzero-weight contraction in Proposition 1.6.2 compute its absolute and relative cohomology. $\square$

Thus the full displayed ghost carrier cannot define semi-infinite cohomology at arbitrary $k \neq -2$. Another semi-infinite construction would need another explicitly defined differential and a comparison with this one at the applicable parameters. The vacuum PBW character $\prod_{n \geq 2} (1 - q^n)^{-1}$ counts states by conformal weight. It does not compute the cohomology in the preceding corollary.

1.8 CENTRAL COEFFICIENTS AND MAPS OVER CURVES

1.8.1 Level reflection and central charge

PROPOSITION 1.8.1 (*Affine scalar normalizations*). Fix nonzero h and a positive integer d . For $k \neq -h$, set

$$k' = -k - 2h, \quad a(k) = \frac{(k+h)d}{2h}, \quad c_{\text{Sug}}(k) = \frac{kd}{k+h}.$$

Then

$$a(k) + a(k') = 0, \quad c_{\text{Sug}}(k) + c_{\text{Sug}}(k') = 2d.$$

For an affine Lie algebra one takes $h = h^\vee$ and $d = \dim \mathfrak{g}$. The half-central-charge normalization has sum d , rather than zero.

Proof. Write $t = k + h \neq 0$. Reflection sends t to $-t$. The two expressions are $a = td/(2h)$ and $c_{\text{Sug}} = d - hd/t$. Their reflected sums give the formulas. $\square$

PROPOSITION 1.8.2 (*Principal central-charge reflection*). Fix rank r and vectors $\rho, \rho^\vee$ in a real inner-product space. For $t \in \mathbb{C}^*$ define

$$c(t) = r - 12(t\|\rho^\vee\|^2 - 2(\rho, \rho^\vee) + t^{-1}\|\rho\|^2).$$

This is the principal central-charge normalization with $t = k + h^\vee$ and long roots of square length 2. For a fixed scalar v , set $\kappa(t) = vc(t)$. Then

$$c(t) + c(-t) = 2r + 48(\rho, \rho^\vee), \quad \kappa(t) + \kappa(-t) = v(2r + 48(\rho, \rho^\vee)).$$

For $\mathfrak{sl}_2$, $c(t) = 13 - 6(t + t^{-1})$ and $v = 1/2$, so the sums are 26 and 13. For $\mathfrak{sl}_3$, $c(t) = 50 - 24(t + t^{-1})$ and $v = 5/6$, so the sums are 100 and $250/3$.

Proof. The t and t^{-1} terms cancel under reflection. For $\mathfrak{sl}_2$, the positive root α satisfies $(\alpha, \alpha) = 2$, and $\rho = \rho^\vee = \alpha/2$ has square $1/2$. For $\mathfrak{sl}_3$, with simple roots α_1, α_2 of square 2 and $(\alpha_1, \alpha_2) = -1$, one has $\rho = \rho^\vee = \alpha_1 + \alpha_2$ of square 2. Substitution proves the two cases. $\square$

The parameter reflection $t \mapsto -t$ differs from Feigin–Frenkel duality, whose dual parameters satisfy $t t^\vee = 1/r^\vee$, where $r^\vee$ is the lacing number [1, p. 566, footnote 1]. For example, $k = -8/3$ reflects to $k' = -4/3$ in $\mathfrak{sl}_2$. Their central charges are 26, 0, and their half-central charges sum to 13. String ghosts contribute central charge -26 , or half-central charge -13 . They cancel the reflected pair's total central charge. For a single matter module, Theorem 1.3.2 instead gives the operator coefficient $(C - 26)/12$. These are distinct scalar normalizations attached to specified expressions.

1.8.2 The equation for transport of an anomaly operator

PROPOSITION 1.8.3 (*Transport of a square*). Let d_X, d_Y be degree-one operators on graded vector spaces and $F : X \rightarrow Y$ a degree-zero map. If $d_Y F = F d_X$, then

$$d_Y^2 F = F d_X^2.$$

In particular, a chain isomorphism between square-zero complexes supplies no identification with a nonzero anomaly operator on another carrier. If F is injective and $d_Y^2 = 0$, its intertwining equation forces $d_X^2 = 0$.

Proof. Apply d_Y to the intertwining equation and use that equation once more. For an injective F , the resulting identity $F d_X^2 = 0$ implies $d_X^2 = 0$. $\square$

PROPOSITION 1.8.4 (*Horizontal comparison*). Let X be a smooth manifold. Let $(E^\bullet, d_E, \nabla_E)$ and $(F^\bullet, d_F, \nabla_F)$ be bounded complexes of finite-rank complex vector bundles with flat connections. Assume d_E, d_F commute with the connections. On $\Omega^p(X, E^q)$ use total differential $\nabla_E + (-1)^p d_E$, and use the same convention for F . A degree-zero bundle map $\phi : E^\bullet \rightarrow F^\bullet$ induces a map of these total complexes if and only if

$$d_F \phi = \phi d_E, \quad \nabla_F \phi = (\mathbf{1} \otimes \phi) \nabla_E.$$

If these identities hold and ϕ is a quasi-isomorphism on each fiber, it induces an isomorphism on the hypercohomology of the two local-system complexes.

Proof. On a section of form degree zero, the two components of the total chain equation lie in different form degrees. They give exactly the displayed identities. The Leibniz rule extends these identities to arbitrary forms. Flat local frames identify each stalk of the kernel local systems with its fiber. The cone of the induced local-system map is therefore acyclic on every stalk. Its cohomology sheaves vanish, so it is zero in the derived category of sheaves. Applying derived global sections gives the asserted hypercohomology isomorphism. $\square$

Example 1.8.5 (*A genus-one obstruction to fiberwise transport*). On the oriented torus $T^2 = \mathbb{R}^2 / \mathbb{Z}^2$, let L_λ be the rank-one complex local system with monodromy $\lambda \in \mathbb{C}^*$ on the first circle and $\mathbf{1}$ on the second. Its cellular cochain complex, with the product orientation, is

$$(\Lambda^\bullet \mathbb{C}^2, (\lambda - \mathbf{1})e_1 \wedge -).$$

This is the tensor product of the two circle complexes with differentials $\lambda - \mathbf{1}$ and $\mathbf{0}$. If $\lambda \neq \mathbf{1}$, contraction by the dual vector e_1^* , divided by $\lambda - \mathbf{1}$, gives $dh + hd = \mathbf{1}$. Thus $H^*(T^2, L_\lambda) = \mathbf{0}$. For $\lambda = \mathbf{1}$, the differential is zero and the dimensions are $(\mathbf{1}, 2, \mathbf{1})$. The fibers in both cases are one-dimensional complexes concentrated in degree zero. No fiberwise dimension or scalar central coefficient determines the genus-one cohomology.

A comparison over the moduli of curves needs maps of its sheaves or local systems, compatible with the connections and with clutching at the boundary. The genus-zero filtered lifting criterion alone does not define such maps. The same requirement applies separately to pairings, anomaly operators, and the products used in a Maurer–Cartan equation.

1.9 INSERTION INTEGRALS AND FLATNESS EQUATIONS

1.9.1 Marked curves and insertion dependence

For $n \geq 3$, let $C_n(\mathbb{P}^1)$ be the space of ordered distinct points on the complex projective line. The diagonal action of $\mathrm{PGL}_2(\mathbb{C})$ is free. There is a unique projective transformation taking the first three points to $\mathbf{0}, \mathbf{1}, \infty$. The remaining points give the explicit quotient

$$C_n(\mathbb{P}^1) / \mathrm{PGL}_2(\mathbb{C}) \cong \mathcal{M}_{\mathbf{0}, n}.$$

Thus the two spaces have complex dimensions n and $n - 3$. Their compactifications retain these dimensions. In particular, a compactification of $C_n(\mathbb{P}^1)$ cannot equal $\overline{\mathcal{M}}_{\mathbf{0}, n+1}$, whose dimension is $n - 2$.

PROPOSITION 1.9.1 (*Insertion integrals and Euler characteristics*). Put $Z = \overline{\mathcal{M}}_{0,n}(\mathbb{C})$ with its complex orientation and real dimension $m = 2n - 6$. Let $C_1, \dots, C_n$ be complex cochain complexes. An insertion map is a degree-zero chain map

$$\mu : C_1 \otimes \cdots \otimes C_n \longrightarrow \Omega^\bullet(Z; \mathbb{C}),$$

where the tensor differential has sign $(-1)^{|x_1| + \cdots + |x_{j-1}|}$ on its j th summand. For closed homogeneous inputs of total degree m , define

$$\mathcal{A}_\mu(x_1, \dots, x_n) = \int_Z \mu(x_1 \otimes \cdots \otimes x_n).$$

This is a multilinear function of their cohomology classes. If, on a fixed product of input spaces, it equals an input-independent scalar for every tuple, that scalar is zero. Therefore an identity with an Euler characteristic of a fixed complex requires separate insertion normalization, and cannot hold for arbitrary insertions when that Euler characteristic is nonzero.

Proof. The chain identity for μ sends a tensor boundary to an exact form. Stokes' theorem on the compact oriented manifold Z makes the integral of such a form zero. Multilinearity follows from the linearity of μ and integration. Putting one input equal to zero forces the scalar to be zero. For an explicit instance, take $n = 3$, Z a point, and $C_j = \mathbb{C}$ in degree zero. The multiplication map gives $\mathcal{A}_\mu(a, b, c) = abc$. The Euler characteristic of the degree-zero complex $\mathbb{C}$ is 1. The tuples $(1, 1, 1)$ and $(0, 1, 1)$ give different amplitudes for the same complex. $\square$

For a chiral-chain application, μ includes the actual insertion and collision maps into forms on Z . Logarithmic forms along a divisor require an integration domain and boundary prescription for which the integration functional is closed. An Euler class integral supplies a specific characteristic form, rather than an arbitrary multilinear insertion.

1.9.2 The differential graded Lie algebra of connections

THEOREM 1.9.2 (*Flatness and the Chern–Simons functional*). Let M be a closed oriented smooth three-manifold and $\mathfrak{g}$ a finite-dimensional complex Lie algebra with a nondegenerate invariant symmetric bilinear form $\langle -, - \rangle$. Consider connections on the trivial bundle, written as $A \in \Omega^1(M; \mathfrak{g})$. On $L = \Omega^\bullet(M) \otimes \mathfrak{g}$ use differential d_{dR} and bracket

$$[\alpha \otimes x, \beta \otimes y] = (\alpha \wedge \beta) \otimes [x, y].$$

For $k \in \mathbb{C}^*$, put

$$S_k(A) = \frac{k}{4\pi} \int_M \left(\langle A \wedge dA \rangle + \frac{1}{3} \langle A \wedge [A, A] \rangle \right).$$

Its critical points are exactly the Maurer–Cartan elements of L :

$$F_A = dA + \frac{1}{2} [A, A] = 0.$$

Proof. The de Rham Leibniz rule and the Lie Jacobi identity make L a differential graded Lie algebra. Its Maurer–Cartan equation in degree one is the displayed equation by definition. For an arbitrary variation $a \in \Omega^1(M; \mathfrak{g})$, integration by parts on the closed manifold gives

$$\left. \frac{d}{dt} \right|_{t=0} \int_M \langle (A + ta) \wedge d(A + ta) \rangle = 2 \int_M \langle a \wedge dA \rangle.$$

Invariance of the bilinear form makes the three differentiated factors of the cubic term equal after the graded signs are included. Its variation is therefore $\int_M \langle a \wedge [A, A] \rangle$. Consequently

$$(dS_k)_A(a) = \frac{k}{2\pi} \int_M \langle a \wedge F_A \rangle.$$

If this vanishes for all smooth variations supported in a coordinate chart, nondegeneracy of the form forces every coefficient of F_A to vanish. The converse follows from the same formula. $\square$

This construction uses $\Omega^\bullet(M) \otimes \mathfrak{g}$. The Chevalley–Eilenberg cochains $\text{Hom}(\Lambda^\bullet \mathfrak{g}, \mathbb{C})$ are a different complex. Transport from a chiral bar object to this differential graded Lie algebra requires the corresponding bracket data or higher morphism components.

1.9.3 Transport of Maurer–Cartan elements

PROPOSITION 1.9.3 (*The components required for Maurer–Cartan transport*). Let L, L' be uncurved L_∞ algebras over $\mathbb{C}$. In the symmetric convention, these are graded vector spaces with degree-one square-zero coderivations Q, Q' on the coaugmented symmetric coalgebras of $L[1], L'[1]$, vanishing on the coaugmentation. Let $\mathfrak{m}$ be the nilpotent maximal ideal of a commutative local Artinian complex algebra. Suppose a coaugmentation-preserving coalgebra map F satisfies $Q'F = FQ$. Writing its Taylor components as $F_j : S^j(L[1]) \rightarrow L'[1]$, define, for $\alpha \in (L \otimes \mathfrak{m})^1$,

$$s\beta = \sum_{j \geq 1} \frac{1}{j!} F_j((s\alpha)^{\odot j}).$$

Then α Maurer–Cartan implies β Maurer–Cartan. A linear chain quasi-isomorphism without these coalgebra equations need not have this property.

Proof. Nilpotence of $\mathfrak{m}$ makes every displayed sum finite. Set $v = s\alpha$ of degree zero and use $e^v = \sum_{j \geq 0} v^{\odot j} / j!$. The coderivation identity gives

$$Q(e^v) = e^v \odot \left(\sum_{j \geq 1} \frac{1}{j!} Q_j(v^{\odot j}) \right).$$

This can be checked by selecting the j inputs on which Q_j acts. The binomial coefficient cancels the factorials. The expression in parentheses is the shifted Maurer–Cartan expression. The coalgebra identity similarly gives $F(e^v) = e^{s\beta}$. Thus $Q(e^v) = 0$ implies $Q'(e^{s\beta}) = Q'F(e^v) = FQ(e^v) = 0$. Projection to $L'[1]$ proves the assertion.

For the last statement, let both underlying complexes have basis x in degree one and y in degree two, with zero differential. Give the first algebra zero brackets, and the second the sole nonzero bracket $[x, x] = y$. Graded skew symmetry permits this bracket because x is odd, and Jacobi holds because y is central. The identity of complexes is a quasi-isomorphism. Over $\mathbb{C}[t]/(t^3)$, tx is Maurer–Cartan in the first algebra but has curvature $t^2 y/2 \neq 0$ in the second. $\square$

COROLLARY 1.9.4 (*Flatness on an identified carrier*). In the preceding proposition, suppose the target is $L' = \Omega^\bullet(M) \otimes \mathfrak{g}$ from Theorem 1.9.2. Then the constructed β is a flat connection with coefficients in $\mathfrak{m}$:

$$d\beta + \frac{1}{2}[\beta, \beta] = 0.$$

The higher Jacobi identities alone do not make every degree-one element Maurer–Cartan.

Proof. The first assertion is the preceding proposition and the definition of curvature on the target. For the second, the abelian differential graded Lie algebra with $L^1 = \mathbb{C}x$, $L^2 = \mathbb{C}y$, and $dx = y$ satisfies all Jacobi identities. The element tx over $\mathbb{C}[t]/(t^2)$ has nonzero Maurer–Cartan expression ty . $\square$

Reduction to a Drinfeld–Sokolov constraint requires the reduction maps and their compatibility with these higher components. The polynomial reduction of Theorem 1.7.1 supplies a chain algebra comparison on its polynomial carrier. Its projection fails Poisson compatibility, so it does not by itself give this transport of flatness equations.

1.10 CHARACTERS AND ELLIPTIC KERNELS

PROPOSITION 1.10.1 (*Translation of an integrally graded character*). Let $V = \bigoplus_{n \geq 0} V_n$ have finite-dimensional components and let $C \in \mathbb{C}$. Suppose $\sum_n \dim(V_n)q^n$ converges for $|q| < 1$. For $\tau \in \mathbb{H}$ put

$$Z_V(\tau) = \sum_{n \geq 0} \dim(V_n) \exp(2\pi i \tau (n - C/24)).$$

Then $Z_V(\tau + 1) = e^{-2\pi i C/24} Z_V(\tau)$. For two such nonzero characters of real central charges $C, \tilde{C}$, the product $Z_V \overline{Z_{\tilde{V}}}$ is invariant under translation if and only if $C - \tilde{C} \in 24\mathbb{Z}$. This translation condition alone does not imply invariance under all modular transformations.

Proof. Each summand acquires the factor $e^{2\pi i n} e^{-2\pi i C/24} = e^{-2\pi i C/24}$. Absolute convergence permits the factor to be taken outside the sum. The conjugate character supplies the inverse phase with $\tilde{C}$, giving the stated condition. Translation is only one generator of the modular group. For example $e^{2\pi i \tau}$ is translation invariant but has different values at $2i$ and $i/2$. $\square$

THEOREM 1.10.2 (*The Heisenberg vacuum character*). The rank-one Heisenberg vacuum module has character

$$Z_H(\tau) = e^{-\pi i \tau / 12} \prod_{n \geq 1} (1 - e^{2\pi i n \tau})^{-1} = \eta(\tau)^{-1}.$$

It satisfies

$$Z_H(\tau + 1) = e^{-\pi i / 12} Z_H(\tau), \quad Z_H(-1/\tau) = (-i\tau)^{-1/2} Z_H(\tau).$$

The square root is the holomorphic branch positive when τ is positive imaginary. The modular weight of this character is $-1/2$, with the inverse eta multiplier.

Proof. A vacuum basis consists of $\prod_{n \geq 1} a_{-n}^{m_n} \mathbf{1}$, with finitely many nonzero $m_n \geq 0$ and conformal weight $\sum n m_n$. Summing the geometric series for each oscillator gives the displayed product. It converges normally on compact subsets of $\mathbb{H}$ because $\sum |q|^n / (1 - |q|^n)$ converges there. The vacuum energy is $-1/24$ since the central charge is one. The translation identity follows directly from the product. The eta transformation formula gives $\eta(-1/\tau) = (-i\tau)^{1/2} \eta(\tau)$ [2, Eq. 23.18.5]. Taking reciprocals proves the second identity with the same branch. $\square$

For the universal Virasoro vacuum, replace the oscillator product by $\prod_{n \geq 2} (1 - q^n)^{-1}$ and the exponent by $-C/24$. Its translation multiplier remains $e^{-2\pi i C/24}$. For an integrally graded affine vacuum with its Sugawara conformal vector, that multiplier uses $C = k \dim \mathfrak{g} / (k + h^\vee)$. Neither computation asserts a one-dimensional modular representation for the whole character space.

PROPOSITION 1.10.3 (*Invariant pairings of character vectors*). Let $\chi : \mathbb{H} \rightarrow \mathbb{C}^r$ satisfy $\chi(\gamma\tau) = \rho(\gamma)\chi(\tau)$ for a specified representation ρ of $SL_2(\mathbb{Z})$. If a constant Hermitian matrix H satisfies $\rho(\gamma)^* H \rho(\gamma) = H$ for every γ , then $\chi(\tau)^* H \chi(\tau)$ is modular invariant.

Proof. Substitution gives $\chi(\gamma\tau)^* H \chi(\gamma\tau) = \chi(\tau)^* \rho(\gamma)^* H \rho(\gamma) \chi(\tau) = \chi(\tau)^* H \chi(\tau)$. $\square$

1.10.1 The Laurent normalization of a meromorphic kernel

For $\Lambda_\tau = \mathbb{Z} + \tau\mathbb{Z}$, define

$$\zeta_\tau(z) = \frac{1}{z} + \sum_{\omega \in \Lambda_\tau \setminus \{0\}} \left(\frac{1}{z - \omega} + \frac{1}{\omega} + \frac{z}{\omega^2} \right), \quad z \notin \Lambda_\tau.$$

The grouped summand is $O(|\omega|^{-3})$ on compact subsets away from the poles, so the series converges normally. Set $E_2 = (12/\pi i) \partial_\tau \log \eta$ and $G_\tau(z) = \zeta_\tau(z) + (\pi^2/3) E_2(\tau) z$.

PROPOSITION 1.10.4 (*Transformation of the normalized kernel*). For $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$, put $j = c\tau + d$. Then

$$G_{\gamma\tau}(z/j) = j G_\tau(z) - 2\pi i c z.$$

Both sides have residue j as functions of z at zero. The pulled-back one-form satisfies

$$G_{\gamma\tau}(z/j) d(z/j) = G_\tau(z) dz - \frac{2\pi i c}{j} z dz,$$

where the differential is in the fiber coordinate z with τ fixed.

Proof. The identity $\Lambda_{\gamma\tau} = j^{-1}\Lambda_\tau$ and reindexing of the normally convergent series give $\zeta_{\gamma\tau}(z/j) = j \zeta_\tau(z)$. The eta multiplier is independent of τ . Logarithmic differentiation of its transformation law [2, Eq. 23.18.5] therefore gives

$$E_2(\gamma\tau) = j^2 E_2(\tau) + \frac{6c j}{\pi i}.$$

For $c = 0$ the same identity follows directly from translation and the central element $-I$. Substitution proves the kernel identity because $2\pi/i = -2\pi i$. The leading term $1/z$ proves the residue assertion, and division by j gives the one-form identity. $\square$

The function G_τ is a meromorphic expression on the covering plane with the stated transformation law. A propagator on an elliptic curve requires its additional periodicity, coefficient bundle, and differential equation. The character, this kernel, and a chiral chain complex are separate objects.

Conjecture 1.10.5 (*Chiral realization of modular transformation laws*). For a conformal chiral algebra with a specified convergent character space and a compatible family of elliptic chiral chain complexes, there is a modular-equivariant chain-level realization of its character transformation law. Such a realization includes descent maps on the family of elliptic curves and a trace map compatible with those descent maps. The scalar character identities above do not construct these maps.

I.II REDUCTION PARAMETERS AND FINITE BATALIN–VILKOVISKY COMPLEXES

PROPOSITION I.II.1 (*Conformal obstruction to parameter reflection*). Let $t \in \mathbb{C}^*$ and put $c(t) = 13 - 6(t + t^{-1})$. A conformal vertex algebra isomorphism between the universal Virasoro algebras at charges $c(t)$ and $c(-t)$ exists if and only if $t = \pm i$.

Proof. A conformal isomorphism preserves the vacuum and conformal vector T . It therefore preserves $T_{(3)}T = (c/2)\mathbf{1}$, so the two charges must agree. Since their sum is 26, equality is equivalent to $c(t) = 13$, or $t + t^{-1} = 0$. This gives $t = \pm i$. Conversely both universal algebras then have charge 13, and the map sending their conformal generators to one another is an isomorphism by their identical universal presentation. $\square$

At $t = -2/3$, reflection gives $c(t) = 26$ and $c(-t) = 0$, whereas the reciprocal partner $t^{-1} = -3/2$ has charge 26. This distinguishes reflection from conformal Feigin–Frenkel duality by an actual vertex product. It does not exclude a curved or nonconformal comparison with different structure-preservation requirements.

Example I.II.2 (*The nonsimply laced central coefficient*). For B_2 , take an orthonormal basis e_1, e_2 and positive roots $e_1 - e_2, e_2, e_1, e_1 + e_2$. The coroot of α is $2\alpha/(\alpha, \alpha)$. Thus

$$\rho = (3/2, 1/2), \quad \rho^\vee = (2, 1), \quad (\rho, \rho^\vee) = 7/2.$$

The principal charge is $c(t) = 86 - 60t - 30/t$, so $c(t) + c(-t) = 172$. The rank is 2, the dimension is $2 + 2 \cdot 4 = 10$, and $h^\vee = 1 + (\rho, e_1 + e_2) = 3$. Consequently $2r + 4h^\vee \dim \mathfrak{g} = 124$ cannot be the reflected sum for all simple Lie algebras. For simply laced root systems $\rho = \rho^\vee$; the general expression remains $2r + 48(\rho, \rho^\vee)$.

PROPOSITION I.II.3 (*Two genus-one scalar normalizations*). Let $B_2 = 1/6$ be the second Bernoulli number and define the scalar coefficient

$$F_1(\kappa) = \kappa \frac{2^1 - 1}{2^1} \frac{|B_2|}{2!}.$$

Then $F_1(\kappa) = \kappa/24$. Under $\kappa = C/2$, this is $C/48$. The vacuum-energy exponent in $q^{L_0 - C/24}$ is instead $-C/24$. At $C = 26$ these values are $13/24$ and $-13/12$.

Proof. The Bernoulli coefficient is $(1/2)(1/6)(1/2) = 1/24$. Substitution gives the other values. $\square$

This proposition concerns the specified Bernoulli coefficient. Its identification with a Hodge integral or a trace requires that map and its normalization. The vacuum energy uses the separately defined conformal trace.

THEOREM I.II.4 (*A polynomial Batalin–Vilkovisky differential and its quotient*). Let $R = \mathbb{C}[x_1, \dots, x_n]$ and $S \in R$. Give $K = R \otimes \Lambda(\eta_1, \dots, \eta_n)$ degrees $|x_i| = 0, |\eta_i| = -1$. Use the degree-one odd Poisson bracket with $\{x_i, \eta_j\} = \delta_{ij}, \{x_i, x_j\} = \{\eta_i, \eta_j\} = 0$, extended by graded skew symmetry and the graded Leibniz rules. Then $Q = \{S, -\}$ is the Koszul differential

$$Qx_i = 0, \quad Q\eta_i = \partial_i S,$$

and satisfies $Q^2 = 0$. If $\partial_1 S, \dots, \partial_n S$ is a regular sequence, the chain algebra projection

$$p : K \longrightarrow R/(\partial_1 S, \dots, \partial_n S), \quad p(\eta_i) = 0$$

is a quasi-isomorphism.

Proof. The bracket formulas give the displayed derivation. Its square vanishes on every generator and hence everywhere. The projection is a chain map because it kills each $Q\eta_i$. For a single regular element f , the complex $R\eta \xrightarrow{f} R$ has zero cohomology in degree -1 and quotient $R/(f)$ in degree zero. Inductively tensor with the next two-term complex. The two-column spectral sequence first computes the preceding Koszul cohomology, then multiplication by the next element on the quotient. Regularity makes that multiplication injective. The filtration is finite, so induction proves concentration in degree zero and the asserted projection isomorphism. $\square$

For $S = \frac{1}{2} \sum_i x_i^2$, the target is $\mathbb{C}$ and the projection evaluates $x_i = \eta_i = 0$. This is an explicit finite polynomial BRST comparison. It has no collision sheaves, ultraviolet completion, or curvewise descent. A comparison with chiral chains must construct those additional objects and maps.

1.12 FORM DEGREE IN HIGHER INTERACTION TERMS

PROPOSITION 1.12.1 (*Quartic expressions on a three-manifold*). Let M be a smooth three-manifold and $A \in \Omega^1(M; W)$ for a finite-dimensional vector space W . Any expression obtained by applying a four-linear map to coefficients of A and wedging all four one-form factors is zero. If a vector field v and a four-linear map $T : W^{\otimes 4} \rightarrow \mathbb{C}$ are specified, the expression

$$T(\iota_v A, A, A, A) \in \Omega^3(M)$$

is instead a defined three-form. It can be nonzero, but depends on the additional contraction and tensor.

Proof. The first expression belongs to $\Omega^4(M) = 0$. For the second, $\iota_v A$ has form degree zero, so the total degree is three. For an explicit local example take $W = \mathbb{C}^3$, $A = e_1 dx + e_2 dy + e_3 dz$, $v = \partial_x$, and the tensor whose only nonzero coordinate entry is $T(e_1, e_1, e_2, e_3) = 1$. Then the expression is $dx \wedge dy \wedge dz$. $\square$

An RTT relation is an algebraic relation on matrix-valued series. To use it as an action term, a map to a three-form such as the displayed contraction must be defined and its required symmetries proved. A critical affine level alone does not make the Lie bracket in $\Omega^\bullet(M) \otimes \mathfrak{g}$ vanish. For example, $A = e dx + f dy$ in $\mathfrak{sl}_2$ has $[A, A]/2 = h dx \wedge dy \neq 0$, independently of any level parameter.

PROPOSITION 1.12.2 (*Integration on a finite quotient*). Let a finite group Γ act freely by orientation-preserving diffeomorphisms on a closed oriented three-manifold M . Let ω be an invariant three-form and $\bar{\omega}$ its descended form on M/Γ . Then

$$\int_{M/\Gamma} \bar{\omega} = \frac{1}{|\Gamma|} \int_M \omega.$$

Proof. The quotient map is an oriented covering of degree $|\Gamma|$. An evenly covered coordinate cover and a partition of unity reduce integration to the sum over its $|\Gamma|$ sheets. Each sheet has the same integral, giving the formula. $\square$

For a nonfree finite action, integration on the quotient stack uses the right-hand side as its normalization. An additional division by $|\Gamma|$ after taking this quotient integral changes the normalization. Neither a fractional grading nor a W-algebra specifies an action of Γ on M .

BIBLIOGRAPHY

- [1] T. Arakawa, *Rationality of W -algebras: principal nilpotent cases*, Annals of Mathematics **182** (2015), 565–604.
- [2] NIST Digital Library of Mathematical Functions, §23.18, Eq. 23.18.5, <https://dlmf.nist.gov/23.18.E5>.